Flight data exploration (CD Project Part 1)

Janika Correia
2119122
Data Science
Faculty of Exact Sciences and Engineering
University of Madeira
Funchal, Portugal

David Abreu
2121422
Data Science
Faculty of Exact Sciences and Engineering
University of Madeira
Funchal, Portugal

***Abstract*—This paper analyses the data of flights delay and cancellation. By extracting useful information using methods within the scope of data science, we identify patterns and form conclusions regarding data.**

I. Introduction

For large corporations that conduct services at a large scale on a daily bases, having data relating to these activities are crucial. When it comes to the aviation industry, ensuring that daily operations are running efficiently is very important not only from a lucrative standpoint but also safety standpoint.

In this project for Data Science, we have have been tasked with analyzing flight delays and cancellation data and to derive interpretations of this data. As well as to identify potential patterns and validate potential model usage. This report details our process in the 1st 3 phases of the data analysis cycle which are: Problem definition, Data analysis and Cleansing as well as initial model Selection.

II. Problem Formulation (phase 1):

A. Problem Definition

The aviation industry faces significant financial and operational losses due to flight disruptions. Leaving passengers frustrated due to lack of effective services, these corporations face issues which leads to increased costs and logistical overhead to combat this. This project addresses the unpredictability of flight delays and cancellations by making use of historical flight data to identify possible causes and provide insight to respond to these problems.

B. Data Analysis Goals

Our objective for this analysis is to gain insight on patterns regarding flight delays. We aim to transform the data provided to use into meaningful and valuable information which can then be used to develop an effective prediction model. Throughout our analysis we will be focusing on Arrival Delay as our central target. More specifically our objectives include: Regression: Determine the impact of certain variables on arrival delay. Classification: Classify the delay type for better determination of the situation. Exploratory objective: Identify operational patterns and group behaviour.

III. Data Analysis and Cleansing (phase 2):

A. Dataset Overview:

The data set has been provided from the U.S. DOT Bureau of Transportation Statistics (2019–2023). This dataset includes all relevant information on the number of on-time, delayed, canceled, and diverted flights. The key features of this dataset include: Categorical identifiers (Airline, Origin Destination), Temporal markers (Schedule and Actual times), Quantitative metrics (Distance and delay time) as well as reasons contributing to the delay (Such as weather, carrier, security, etc.)

B. Data pre-processing and cleansing:

1) Removal of Non-Completed Flights:: Rows where CANCELLED or DIVERTED are 1 have been removed. This is due to the fact that since they have never arrived to a destination, there will not exist an arrival delay which is our target variable. Therefore this information is irrelevant.

2) Target Variable Cleanup:: Rows where arrival delay is null are dropped for the same reason as mentioned above.

3) Normalization of Delays: : Arrival delay is normalized to convert negative delay values (early arrivals) to 0. This is to prevent the data from being skewed and preventing the effectiveness of our model.

4) Outlier Handling:: By targeting DISTANCE and Scheduled elapsed time, we have identified outliers using the Interquartile range. These extreme values are replaced with the mean of the column to ensure extreme anomalies don't distort the model's learning.

5) Mean Imputation:: Any remaining missing numeric values are filled with the mean of their respective columns to maintain a complete dataset without losing more rows.

6) Dataset Balancing: : The dataset has been balanced 50/50 between delayed flights and on-time flights. This is to ensure that our model will have a balanced amount of data that is relevant to the target but also realistic.

7) Redundancy Reduction (ID Stripping): : Redundant identifiers like FL NUMBER, ORIGIN CITY, and DOT CODE have been removed has these columns contain duplicate data which helping the model focus on the most impactful features.

8) Target Classification:: Finally, we transform the continuous delay minutes into a classification where this classifications suggests that:

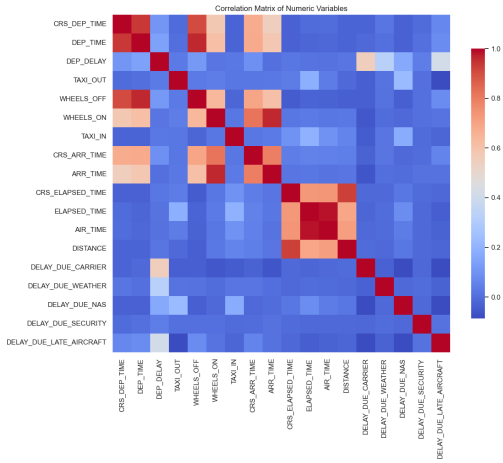

Fig. 1. Correlation matrix

- On-time: < 15 minutes
- Short delay: 15–30 minutes
- Long delay: > 30 minutes.

IV. Exploratory Data Analysis (EDA):

During this phase we performed statisitical analysis on the dataset and derived graphs and statistical information.

A. Descriptive Statistics and Visualizations:

- The histograms and boxplots reveal a dataset with two distinct types of data: Temporal (Clock-based) and Duration (Minute-based).
- Temporal Features: CRS DEP TIME and WHEELS OFF show a "multi-modal" distribution. There are clear peaks in the early morning (around 06:00–08:00) and late afternoon. Flights are rarely scheduled in the middle of the night.
- Target Variable (ARR DELAY): Before balancing, the distribution is extremely "right-skewed." The vast majority of flights cluster near zero, with a very long "tail" representing the rare but extreme delays.
- Class Separation: In the grouped box plots, it can be seen that Long Delays (orange/green) tend to correlate with later departure times in the day compared to On-time flights (blue).

B. Patterns, Outliers, and Correlations:

- Box plots show significant outliers in TAXI OUT and DEP DELAY. Some flights spend over 120 minutes taxiing—these are rare anomalies that would have skewed a linear model if you hadn't implemented the IQR clipping in your
- The Grouped Distributions show that as the day progresses (higher CRS DEP TIME), the proportion of "Long Delays" increases. This suggests a "snowball effect" where delays accumulate throughout the network over the course of a day.

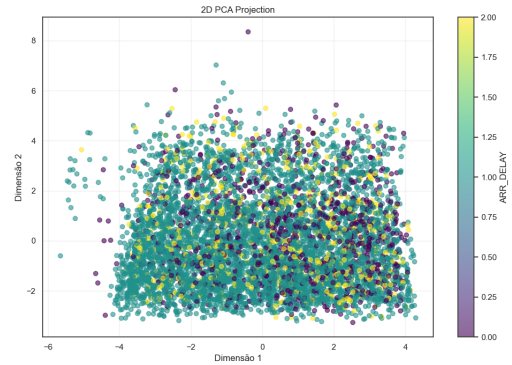

Fig. 2. PCA graph

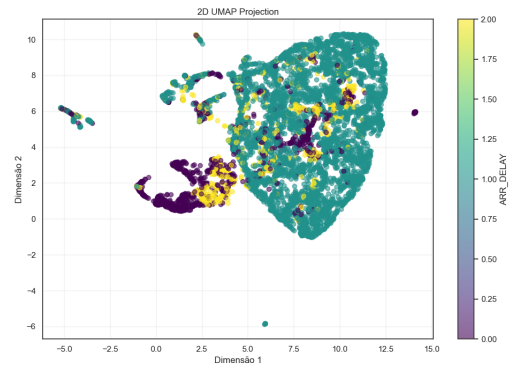

Fig. 3. UMAP graph

- Correlation Analysis From the Correlation Matrix: High Multicollinearity: There is a near-perfect correlation ($r \approx 1.0$) between DISTANCE, AIR TIME, and CRS ELAPSED TIME.Insight: In your final model, you should likely keep only one of these to avoid redundant information.
- Weak Predictors: Interestingly, DISTANCE has almost zero correlation with DEP DELAY. This means a long-haul flight is no more likely to be delayed than a short-hop flight.The "Leakage" Smoking Gun: DEP DELAY and ARR DELAY are very highly correlated.

C. Dimensionality Reduction (PCA vs. UMAP)

1) Linear Method: PCA (2D Projection): When Principal Component Analysis (PCA) was applied to the dataset, the goal was to see if the main drivers of flight delays could be captured through a linear combination of our features. However, as the 2D projection shows, the result is a dense, overlapping 'cloud' where the different delay categories—from on-time to long delays—don't naturally pull apart into distinct groups. This lack of separation identifies that the relationship between a flight's schedule and its eventual delay is **highly complex and non-linear**.

2) Non-Linear Method: UMAP (2D Projection): The UMAP plot is much more revelaing over this dataset,

showing distinct ”islands” and clusters. The ”islands” likely represent specific routes or airport areas. Some clusters are teal which suggest on time flights and others yellow/purple which suggest very delayed. This suggests that the combination of specific origin-destination pairs and airlines shows clear distinctions that the linear model missed.

D. Initial Insights from EDA

The Exploratory Data Analysis confirms that flight delays are not random occurrences, but rather the result of a complex, non-linear system driven by two primary factors: **temporal information** and **location**.
The accumulated delay highlighted by the shift in histograms and box plots demonstrates that the hour of departure is a critical predictor, as delays worsen as the day progresses. Additionally the cluster effect observed in our UMAP analysis proves that the aviation network performance is heavily influenced by specific carrier-route combinations and airport hubs. Finally, the high multicollinearity between distance and time metrics, alongside the linear failure of the PCA, justifies more reliability toward non-linear machine learning models.

V. Feature Engineering

After an analysis on the raw data has been performed, Feature Engineering can take place. We previously identified that temporal variables are significant and there exists value in using them to extract more information. We have extracted 21 new predictors features which fall into the following categories: Temporal Features, Route and Operational information,

1) 1. Temporal Feature Extraction: To identify trends due to a certain periods in time we have created:

- – MONTH
- – QUARTER
- – DAY OF THE WEEK
- – IS_WEEKEND
- – IS_HOLIDAY_SEASON

2) 2. Operational performance: To identify operational performance of flights we have categorized time periods of a day:

- – DEP HOUR
- – MORNING FLIGHT
- – AFTERNOON FLIGHT
- – NIGHT_FLIGHT These features allows us to evaluate the operational performance in a daily cycle, allowing for deeper inspection of daily operations.

3) Route information: : By combining origin and destination into a **ROUTE** feature and calculating **ROUTE FREQUENCY**, we identified popular routes takes.

4) Performance and Scheduling Metrics:: To help the model handle non-linear relationships, we created several derived categories:

- – PLANNED SPEED
- – **DISTANCE_CAT** (duration category)
- – DURATION_CAT (duration category)
- – SPEED_CAT (speed category)
- – DISTANCE_POW2
- – DISTANCE_x_ELAPSED_TIME

VI. Hypothesis testing:

Finally, to evaluate our earlier insights during EDA, we performed hypothesis testing. This was done to identify if our conclusions drawn are valid and if we can follow through with using them in future phases.

1) **Hypothesis:** Certain airlines are systematically delayed compared to others.
   a) **Null Hypothesis:** The delay pattern (mean) is equal across all airlines.
   b) **Alternative Hypothesis:** At least one airline has a statistically different systematic delay pattern (higher or lower) than the others.
   c) **Test used:** Kruskal-Wallis H-Test was used because `ARR_DELAY` does not follow a normal distribution (it is heavily right-skewed). This test compares the medians of groups to determine if they come from the same distribution.
   d) **Results:** * P-Value: < 0.001
   e) **Interpretation:** Since the p-value < 0.05, we reject the Null Hypothesis. There is strong statistical evidence that certain airlines are systematically more delayed than others.
2) **Hypothesis :** Weather-related factors have a statistically significant impact on flight delays.
   a) **Null Hypothesis:** Weather variables have no relationship with delay times.
   b) **Alternative Hypothesis**: There is a statistically significant relationship between weather intensity and delay duration.
   c) **Test used:** Kruskal-Wallis H-Test was used because `ARR_DELAY` does not follow a normal distribution (it is heavily right-skewed). This test compares the medians of groups to determine if they come from the same distribution.
   d) **Results:** P-Value: $¡ 0.001$.
   e) **Interpretation:** We **Reject the Null Hypothesis**. The presence of a weather flag (`IS_WEATHER_DELAY`) has a statistically significant relationship with the magnitude of the arrival delay.